

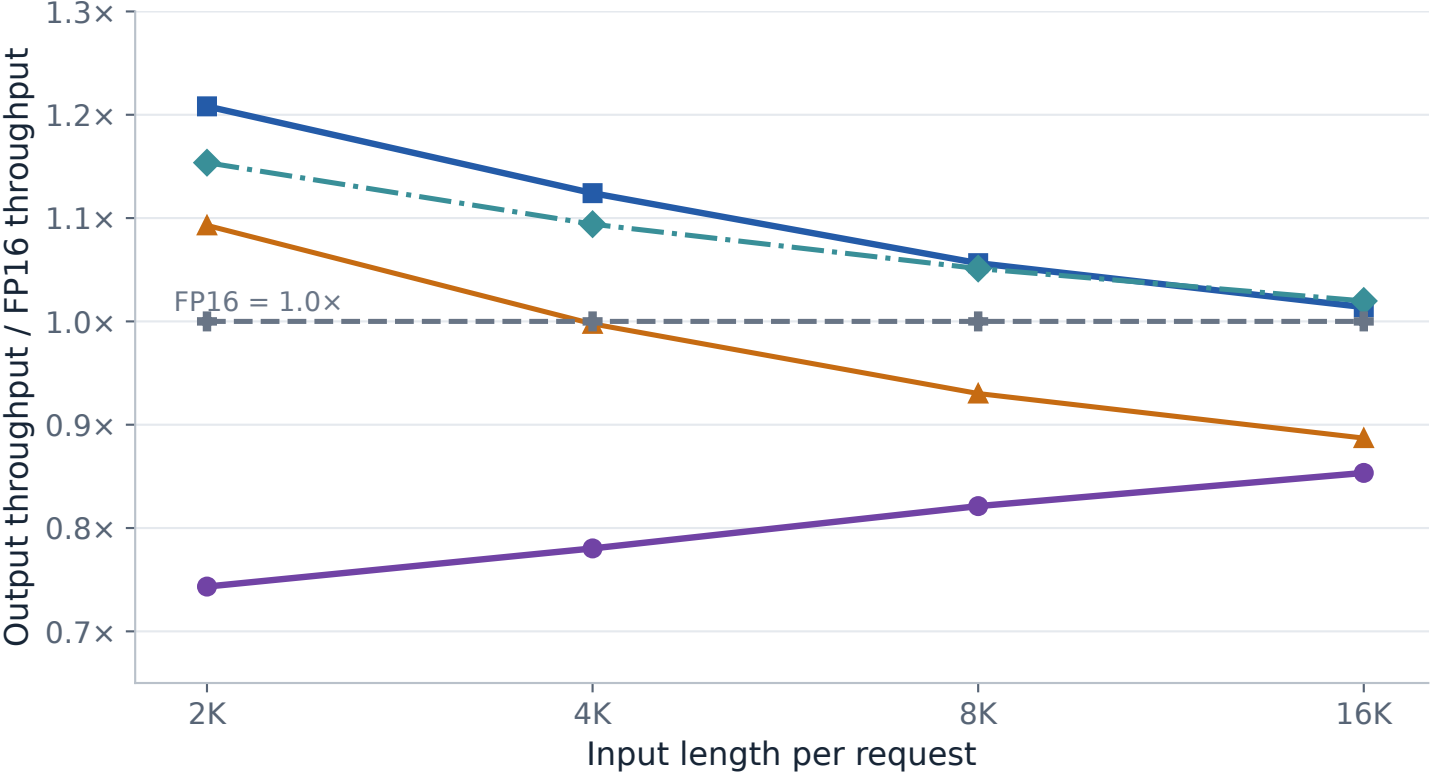
Runtime Cost

Quantization and concurrency across context · two plots, four findings

IQ1_M Q2_K Q4_K_M Q8_0 FP16

Quantization × Context

8B · fixed 32 clients · FP16 baseline at each input length



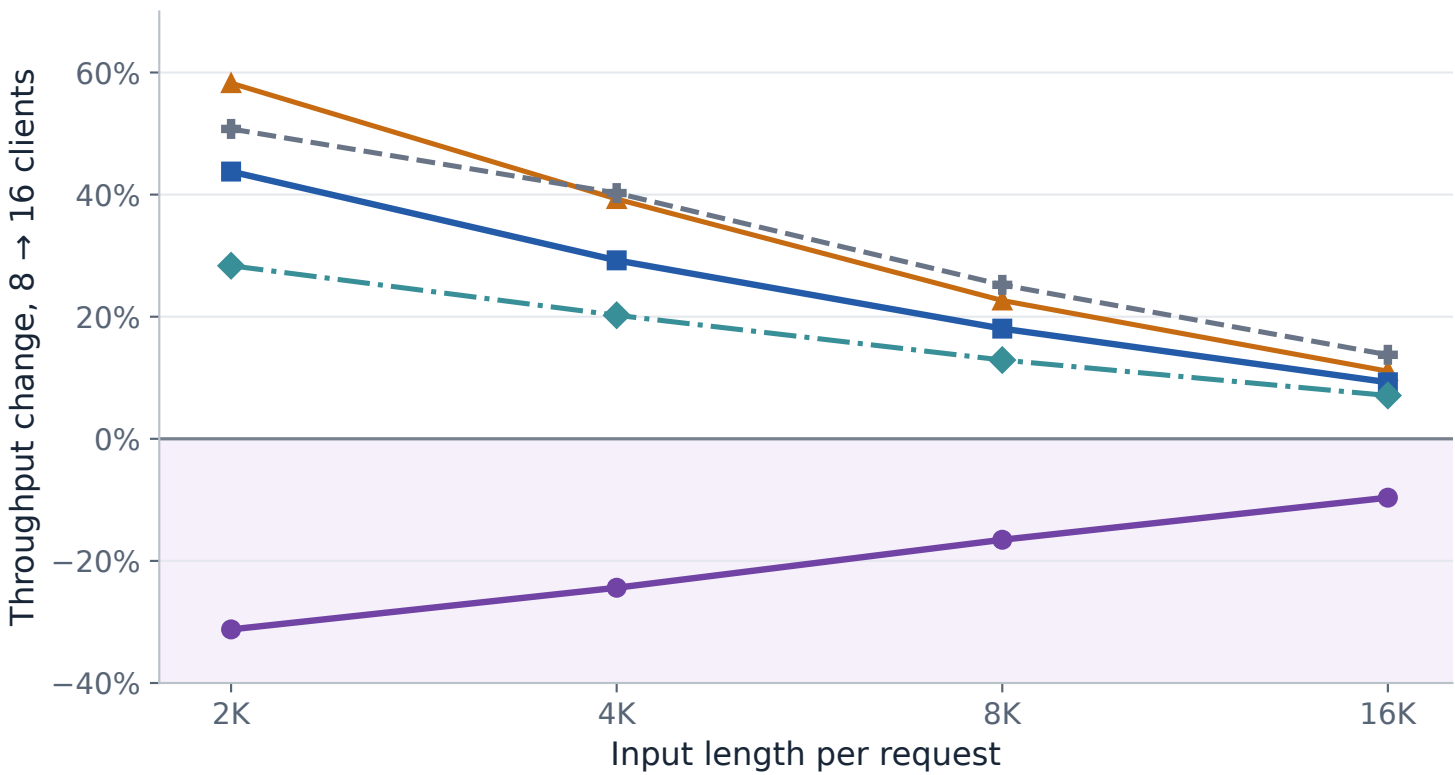
F1 Lower bits ≠ better runtime

F4 Long context erodes quantization gains

Separate 70B result · 2K input · 8 clients
Q2 vs. Q4: +29% TTFT p95, +19% TPOT p95

Concurrency Scaling × Context

8B · throughput change from 8 → 16 clients



F2 More concurrency ≠ more throughput

F3 Long context weakens concurrency scaling

8B IQ1_M @ 2K: dip and recovery
353 → 243 → 350 tok/s (8 → 16 → 32 clients)

Two RTX A6000 GPUs · FP16 KV · 512 outputs/request · prompt reuse off · K = 1,024 input tokens
Measured points: one run per setting; thermal limiting occurred in some settings.